

Math 121: Commutative Algebra

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1 Rings

Important Example: Let X be a “geometric set”, and consider the ring

$$\text{Hom}(X, \mathbb{R}) = \{X \xrightarrow{f} \mathbb{R}\}$$

with pointwise addition and multiplication

$$(f + g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x)$$

$r \in \text{Hom}(X, \mathbb{R})$ is a unit if and only if there exists $g \in \text{Hom}(X, \mathbb{R})$ such that $fg = 1$ if and only if $\forall x \in X, (fg)(x) = f(x)g(x) = 1$

$$\text{Hom}(X, \mathbb{R})^\times = \text{Hom}(X, \mathbb{R}^\times)$$

Definition 1.1. Let $a \in R, a \neq 0$

- a is a unit if and only if $\exists b \in R$ such that $ab = 1$,
- a is a zero divisor if and only if $\exists b \in R \setminus \{0\}$ such that $ab = 0$,

Lemma 1.2. The set of units $R^\times = \{a \in R \mid a \text{ is a unit}\}$ is a group under multiplication.

Proof: Worksheet (03/31).

Lemma 1.3 (Useful Lemma). If $\phi : R \rightarrow S$ is a ring homomorphism, then it follows from the definition that

- $\phi(0_R) = 0_S$
- $\phi(-a) = -\phi(a)$
- $\phi(a^{-1}) = \phi(a)^{-1}$

Remark 1.4. $\mathbb{Z}$ is initial in the category of rings. There is a unique homomorphism from the initial object to every other object. For any ring R ,

$$\text{Hom}(\mathbb{Z}, R) = \{1 \rightarrow 1_R\}$$

We don't know anything about homomorphisms to the initial object.

2 Ideals

Lemma 2.1. *If $\phi : R \rightarrow S$ is any ring homomorphism and $J \subset S$ is an ideal $\phi^{-1}(J)$ is also an ideal.*

Proof: Worksheet Q4.1.

Theorem 2.2. *If $I \subset R$ is an ideal then,*

1. *There exists a ring R/I and a surjective ring homomorphism $\pi : R \rightarrow R/I$ such that $\ker(\pi) = I$. Moreover R/I and π are unique upto isomorphism.*
2. *There is an inclusion preserving bijection*

$$\{\text{Ideals in } R/I\} \leftrightarrow \{\text{Ideals in } R \text{ containing } I\}$$

$$\bar{J} \leftrightarrow \pi^{-1}(\bar{J})$$

Proof.

1. Construct R/I explicitly.
2. I gives an equivalence relation $\sim$ on R such that

$$a \sim b \iff a - b \in I \iff a = b + I$$

Show that equivalence classes $R/\sim$ is a ring.

$$R \rightarrow R/I = R/\sim$$

$$a \mapsto \bar{a} = a + I$$

Let $\bar{J}$ be an ideal in R/I , and J be an ideal in R and define the maps

$$\phi : \bar{J} \rightarrow \pi^{-1}(\bar{J}), \quad \psi : J \rightarrow \pi(J)$$

Now we want to show that $\phi \circ \psi = \psi \circ \phi = \text{id}$. Consider $\phi \circ \psi(J)$ and $\psi \circ \phi(\bar{J})$. Note that

$$\psi \circ \phi(\bar{J}) = \pi(\pi^{-1}(\bar{J})) = \bar{J}$$

since π is surjective. Thus $\psi \circ \phi = \text{id}$.

Now, note that $J \subseteq \phi \circ \psi(J) = \pi^{-1}(\pi(J))$, and since $\bar{0} \in \pi(J)$, $I \subseteq \pi^{-1}(\pi(J))$. An element $x \in \pi^{-1}(\bar{J})$ looks like $x + I$, and $\pi^{-1}(\bar{J}) = J + I$. Moreover since $\bar{0} \in \bar{J}$, we have $\pi^{-1}(\bar{0}) \subseteq \pi^{-1}(\bar{J})$ which implies $I \subseteq J$. Thus, $\psi \circ \phi(J) = J$ and $\psi \circ \phi = \text{id}$.

□

Exercise: Describe the ideals in

1. $\mathbb{Z}/\langle 24 \rangle = \{\langle 0 \rangle, \langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 4 \rangle, \langle 6 \rangle, \langle 8 \rangle, \langle 12 \rangle\}$
2. $\mathbb{Z}/\langle n \rangle = \{\langle d \rangle : d|n\}$
3. $\mathbb{Z}/\langle 3 \rangle = \{\langle 0 \rangle, \langle 1 \rangle\}$
4. $\mathbb{Z}/\langle 5 \rangle = \{\langle 0 \rangle, \langle 1 \rangle\}$
5. $\mathbb{C}[x, y]/\langle y - x^2 \rangle \cong \mathbb{C}[x] = \{\text{Ideals in } \mathbb{C}[x]\}$
6. $\mathbb{C}[x]/\langle x^n \rangle = \{\langle x^i \rangle : 0 \leq i < n\}$

Lemma 2.3. *If $\phi : R \rightarrow S$ is a ring homomorphism with $\ker(\phi) \supseteq I$ then there exists a unique morphism $\tilde{\phi} : R/I \rightarrow S$*

$$\begin{array}{ccc}
 R & \xrightarrow{\pi} & R/I \\
 \downarrow \phi & \searrow \tilde{\phi} & \\
 S & &
 \end{array}$$

Universal Property of Quotients: *Upto isomorphism R/I is the ring such that*

$$\text{Hom}(R/I, S) = \{\phi : R \rightarrow S : I \subset \ker(\phi)\}$$

We want to define $\tilde{\phi}(\bar{a}) = \tilde{\phi}(a + I) = \phi(a)$ and $\tilde{\phi}(\bar{0}) = 0$.

Lemmact 2.4. *Maximal Ideal $\implies$ Prime Ideals.*

Definition 2.5. *The spectrum of a ring R is the set*

$$\text{Spec}(R) = \{\text{Prime Ideals in } R\}$$

the maximal spectrum of R is the set

$$\mathfrak{m}\text{Spec}(R) = \{\text{Maximal Ideals in } R\}$$

$\text{Spec}(R)$ is where the ‘geometry lives’. Note that $\mathfrak{m}\text{Spec}(R) \subset \text{Spec}(R)$.

Example: $R = \mathbb{Z}[\sqrt{-5}]$. 2 is irreducible but not prime. $R/\langle 2 \rangle = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Example: $R = \mathbb{C}[x, y]/\langle x^3 - y^2 \rangle$. x is irreducible, but $\langle x \rangle$ is not prime. $R/\langle x \rangle = \mathbb{C}[y]/\langle y \rangle$ and y is a zero divisor.

Theorem 2.6. *Let R be a non-zero ring.*

1. R has a maximal ideal.
2. If $I \subsetneq R$ is a proper ideal there exists a prime ideal $P \subseteq R$ such that $I \subset P \subsetneq R$.

Proof. **Zorn's Lemma:** Let $(P, \leq)$ be a po-set.

If $P \neq \emptyset$ and if $C = C_1 \leq C_2 \leq C_3 \leq \dots$ is a chain in $(P, \leq)$ then there exists $B \in P$ such that $C_i \leq B$ for all i , then there exists $M \in P$ such that if $S \leq M$ then $M = S$.

Let $P = \{\text{Proper ideals in } R\}$ and $\leq$ be inclusion. We claim that $(P, \leq)$ is a poset.

1.
 - $P \neq \emptyset$ since $\langle 0 \rangle \in P$.
 - Given a chain in P

$$I_1 \subseteq I_2 \subseteq \dots$$

Then, define $B = \bigcup I_j$. Note that B is an ideal since union of nested ideals is an ideal. And B is proper since $1 \notin I_j$ for all j , and hence $1 \notin B$. Then there exists some ideal $M \subset R$ such that M is maximal.

2. Follows from (1). Let $R = R/I$. Pre-image of prime ideals is prime.

Thus, $\text{mSpec}(R) \neq \emptyset$ and $\text{Spec}(R) \neq \emptyset$. □

3 Zariski Topology

Definition 3.1. If $S \subseteq R$ is any subset of R ,

$$V(S) = \{[P] \in \text{Spec}(R) | S \subseteq P\} \subseteq \text{Spec}(R)$$

Definition 3.2. An algebraic subset of $\mathbb{K}^n$ of some field $\mathbb{K}$ is a set of the form

$$V(\mathcal{S}) = \{P \in \mathbb{K}^n | f(P) = 0, \forall f \in \mathcal{S}\}$$

for a subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$

Examples:

- $V(x^2 + y^2 - 1) \subseteq \mathbb{R}^2$

$$\{(x, y) \in \mathbb{R}^2 | x^2 + y^2 - 1 = 0\}$$

- $\mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{R}^3$

$$\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 - 1 = 0\}$$

- $\mathbb{V}(x^2 - y) \subseteq \mathbb{R}[x, y]$

$$\{(x, y) \in \mathbb{R}[x, y] \mid x^2 = y\}$$

- $\mathbb{V}(x^2 - xy) \subseteq \mathbb{R}[x, y]$

$$\{(x, y) \in \mathbb{R}[x, y] \mid x = 0, x - y = 0\}$$

- $\mathbb{V}(x^2 + 1) \subseteq \mathbb{R}[x]$

$$\mathbb{V}(x^2 + 1) = \emptyset$$

- $\mathbb{V}(x^2 + 1) \subseteq \mathbb{C}[x]$

$$\mathbb{V}(x^2 + 1) = \{i, -i\}$$

- $\mathbb{V}(x^2 + 4) \subseteq \mathbb{F}_3[x]$

$$\mathbb{V}(x^2 + 4) = \emptyset$$

- $\mathbb{V}(x^2 + 4) \subseteq \mathbb{F}_p[x]$

TODO:

- $\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) \subseteq \mathbb{R}[x, y]$

$$\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) = \mathbb{V}(x^3(x - y)^3) = \{(x, y) \in \mathbb{R}[x, y] \mid x = 0, x - y = 0\}$$

- $\mathbb{V}(n(x^2 + y^2 + z^2), n \in \mathbb{Z}) \subseteq \mathbb{R}^3$

TODO:

- $\mathbb{V}(\text{a collection of any linear polynomials in } \mathbb{K}[x_1, \dots, x_n])$

TODO:

Definition 3.3. Let $V \subseteq \mathbb{K}^n$ be an algebraic set. The defining ideal of V is

$$\mathbb{I}(V) = \{f \in \mathbb{K}[x_1, \dots, x_n] \mid f(p) = 0, \forall p \in V\}$$

Definition 3.4. If $\mathfrak{M} \subseteq R$ is a maximal ideal, the residue field of $\mathfrak{M}$ is $R/\mathfrak{M}$.

Definition 3.5. Let K be a field. A K algebra is the data of a ring R together with a fixed ring homomorphism

$$\alpha : K \rightarrow R$$

Definition 3.6. If $(R, K \xrightarrow{\alpha} R)$ and $(S, K \xrightarrow{\beta} S)$ are K -algebras, a ring map $\phi : R \rightarrow S$ is a morphism of K -algebras iff

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ \alpha \uparrow & \searrow \beta & \\ K & & \end{array}$$

Lemma 3.7. If R is k -algebra, R/I has a natural k -algebra structure compatible with the quotient map.

$$\begin{array}{ccc} R & \xrightarrow{\pi} & R/I \\ \alpha \uparrow & \searrow \exists! & \\ k & & \end{array}$$

4 Gröbner basis

Remark 4.1. *Ideal Membership Problem:* Suppose $I \subseteq R[x_1, \dots, x_n]$ an ideal. How can we algorithmically determine if $f \in I$?

Definition 4.2. A ring R is Noetherian if and only if every ascending chain of ideals stabilizes (Ascending Chain Condition), i.e.

$$I_1 \subseteq I_2 \subseteq \dots$$

there exists $N \in \mathbb{Z}$ such that $I_k = I_{k+1}$ for all $k \geq N$.

Lemma 4.3. R is Noetherian if and only if every ideal is finitely generated.

Lemma 4.4 (Dickson's Lemma). Every monomial ideal in $K[x_1, \dots, x_n]$ is finitely generated.

Theorem 4.5 (Hilbert's Basis Theorem). If R is a Noetherian ring, then $R[x]$ is also Noetherian.